

ADVANCED PERSISTENT THREATS

APT GROUP ANALYSIS

TARGET: THE UNITED STATES AND LATIN AMERICA

June 27, 2021

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Abstract

This article examines the Advanced Persistent Threat (APT) groups attributed to the **United States** and **Latin America**. It reviews the principal threat actors associated with these regions, the techniques they employ, and the objectives of their cyber operations.

Keywords: Cybersecurity, Cybercrime, APT, Threat, United States, Latin America

INTRODUCTION

At some point in this series it becomes essential to examine the APT groups attributed to the American continent.

The cyber capabilities attributed to the **United States** and those attributed to **Latin America** differ considerably in both scale and complexity.

While only a handful of APT groups have been publicly associated with countries such as **Brazil** and **Mexico**, the United States has been linked to numerous highly sophisticated cyber operations with global reach.

MEXICO – LEETMX

Mexico has been associated with the **LeetMX** malware campaign, first identified in **2016**.

The campaign relied primarily on malicious email attachments to distribute malware across several countries, including:

- Argentina
- Costa Rica
- El Salvador

- Guatemala
- Mexico
- The United States

According to **ClearSky**, the operation focused primarily on:

- Information theft
- Cyber espionage
- Criminal intelligence gathering

Unlike many state-sponsored APT campaigns, LeetMX appeared to pursue predominantly criminal objectives.

BRAZIL – POSEIDON

In late **2015**, researchers at **Kaspersky** completed their investigation into a Brazilian-attributed APT group known as **Poseidon**.

Evidence suggested that the group had been active since at least **2005**, although malware samples dating back to **2001** were also discovered, indicating a much longer period of development.

Poseidon specialized in attacks against corporate networks.

Its operations generally followed the same pattern:

1. Spear-phishing campaigns delivered malware.
2. Trojans established persistent access.
3. Sensitive corporate information was exfiltrated.
4. Victims were blackmailed.
5. The attackers offered "security consulting services" to resolve the very compromises they had created.

Once hired, the attackers often maintained access to the victim's infrastructure, enabling future espionage or additional compromises.

The malware primarily targeted:

- Windows operating systems.
- Systems configured in English or Portuguese.

One particularly unusual characteristic of Poseidon was its reported use of **hijacked satellite communications** during some operations.

UNITED STATES

At the time this article was written, at least **five APT groups** and one major joint cyber operation had been publicly attributed to the United States.

The joint operation was widely believed to have been conducted in cooperation with Israel.

STRIDER

Strider has reportedly operated since at least **2011**.

The group is associated with malware known as **Remsec**.

Its targets have primarily consisted of organizations and individuals considered valuable to national intelligence services.

Remsec was specifically designed for cyber espionage and includes capabilities such as:

- Keylogging.
- File theft.
- Intelligence collection.
- Long-term persistence.

Researchers found infections in carefully selected organizations, including:

- Institutions located in Russia.
- An airline in China.
- An organization in Sweden.
- An embassy in Belgium.

The limited number of highly specific victims enabled Strider to maintain an exceptionally low operational profile.

EQUATION GROUP

Among all publicly known APT groups, **Equation Group** is widely regarded as one of the most technically sophisticated ever discovered.

Researchers believe the group is linked to the **Tailored Access Operations (TAO)** division of the **U.S. National Security Agency (NSA)**.

TAO's mission is believed to include the collection of foreign intelligence through advanced cyber operations.

Researchers at Kaspersky documented approximately **500 confirmed infections** across **42 countries**, including:

- Iran
- Russia
- Pakistan
- Afghanistan
- India
- Syria
- Mali

Because Equation Group malware incorporated self-destruct mechanisms capable of deleting forensic evidence, researchers believe the true number of infections was significantly higher.

Documents disclosed by **Edward Snowden** suggested that the NSA used **Regin** during operations targeting the Belgian telecommunications provider **Belgacom**.

This raised the possibility that allied intelligence services later exploited the same compromised infrastructure for their own intelligence collection activities.

SHADOW BROKERS

According to several investigations, **Shadow Brokers** appeared to have access to cyber tools developed by the **NSA**.

The group became internationally known after leaking a large collection of offensive cyber capabilities and software exploits believed to belong to **Equation Group**.

The disclosure, which began in **2016**, had far-reaching consequences.

Once these exploits became publicly available, numerous threat actors rapidly incorporated them into their own operations, leading to a dramatic increase in cyberattacks worldwide.

Some of the most damaging malware outbreaks that leveraged these leaked exploits included:

- **WannaCry**
- **NotPetya**

Although the identity of Shadow Brokers has never been conclusively established, the incident remains one of the most significant intelligence leaks in the history of cybersecurity.

The **ThaiCERT** research community later published a detailed technical analysis of the leaked tools and their operational capabilities.

LONGHORN

Longhorn has reportedly been active since at least **2011**.

The group specializes in infiltrating government agencies and international organizations through the use of:

- Custom-developed Trojans.
- Zero-day vulnerabilities.
- Highly targeted intrusion techniques.

Its known targets include organizations operating in:

- Financial services.
- Telecommunications.
- Energy.
- Aerospace.
- Information technology.
- Education.
- Natural resources.

These sectors strongly suggest that Longhorn's activities were intended to support government intelligence collection rather than conventional cybercrime.

The malware attributed to Longhorn was specifically engineered for espionage operations.

Its capabilities include:

- Network reconnaissance.
- Credential collection.
- Data exfiltration.
- Long-term persistence.

Operational security was one of the group's defining characteristics.

The malware communicated with command-and-control infrastructure only at carefully selected intervals, limited the volume of transmitted information, and randomized communication timing to reduce the likelihood of detection during long-term intrusions.

Several of Longhorn's cyber tools were eventually exposed through the **Vault 7** disclosures.

VAULT 7 AND VAULT 8

During **2017**, nearly **10,000 classified documents** describing cyber tools, methodologies, exploits, and operational procedures attributed to **Longhorn**, a group reportedly associated with the **CIA**, were leaked to the public.

The documents were released through **WikiLeaks** under the names:

- **Vault 7**
- **Vault 8**

The disclosures represented one of the largest leaks of offensive cyber capabilities ever made public.

Although many of the exposed vulnerabilities have since been patched, others were subsequently adopted by additional Advanced Persistent Threat groups.

Investigators later identified the individual responsible for the leak as **Joshua Adam Schulte**, a former CIA software engineer, who was arrested in **2018** and later convicted on multiple charges related to the unauthorized disclosure of classified information.

OPERATION OLYMPIC GAMES

Operation Olympic Games is widely regarded as one of the most sophisticated cyber operations ever conducted.

The operation is believed to have been a joint effort between:

- **Equation Group (NSA).**
- **Unit 8200**, the Israeli military's elite signals intelligence and cyber warfare organization.

Previously known as **Unit 515** and later **Unit 848**, Unit 8200 is responsible for:

- Signals Intelligence (SIGINT).
- Codebreaking.
- Cyber operations.
- Electronic intelligence.

The operation reportedly began during the **George W. Bush Administration** in **2007**.

Following the inauguration of **President Barack Obama** in **2009**, the program was expanded and cyber operations accelerated.

In **2010**, however, a programming error allowed the malware to escape the intended target environment.

Instead of remaining confined within Iran's **Natanz Nuclear Facility**, the malware spread across networks around the world, where it was eventually discovered and analyzed by cybersecurity researchers.

The malware became known as **Stuxnet**.

Unlike traditional espionage malware, Stuxnet was designed not only to collect intelligence but also to physically sabotage industrial control systems by disrupting uranium enrichment centrifuges.

For the first time, malware demonstrated the capability to produce measurable physical destruction of critical infrastructure.

Many cybersecurity experts consider Stuxnet the beginning of the modern era of cyber warfare.

CONCLUSION

The Advanced Persistent Threat groups attributed to the United States and Latin America illustrate two markedly different cyber landscapes.

While Latin American operations have generally been fewer in number and often financially motivated, U.S.-attributed groups have been associated with some of the most technically advanced cyber espionage and sabotage campaigns ever documented.

Operations such as **Olympic Games**, together with the capabilities attributed to **Equation Group**, **Longhorn**, and **Strider**, demonstrate the increasing importance of cyberspace as a strategic domain alongside land, sea, air, and space.

At the same time, incidents involving **Shadow Brokers** and **Vault 7** highlight the risks posed when highly sophisticated offensive cyber capabilities are exposed, allowing other threat actors to weaponize them.

Understanding these operations provides valuable insight into the evolution of modern cyber conflict and the growing role of state-sponsored cyber activities in international security.

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